Hum 4147
Technology, Environment and Society

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- Sustainability and Sustainable Development
- Environmental Stewardship
- Environmental Justice and Equity

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Sustainability and Sustainable Development



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Sustainability and Sustainable Development

In 1987, the United Nations World Commission on Environment and Development released the report Our Common Future, commonly called the Brundtland Report.[1] The report included a definition of "sustainable development" which is now widely used:[1][10]

Sustainable development is a development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
It contains two key concepts within it:

https://en.wikipedia.org/wiki/Sustainable_development

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Sustainability and Sustainable Development

The concept of '**needs**', in particular,
the essential needs of the world's poor, to which overriding priority should be given; and

The idea of **limitations** imposed by
the state of technology and social organization on the environment's ability to meet present and future needs.

— World Commission on Environment and Development, Our Common Future (1987)

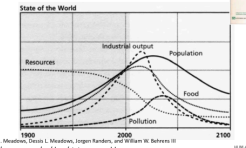
https://en.wikipedia.org/wiki/Sustainable_development

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Sustainability and Sustainable Development

Need for sustainable development issue came after team of scientist from MIT, 1972





Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, and William W. Behrens III
https://www.un.org/en/desa/state-our-world

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Sustainability and Sustainable Development

5 Important Measures for Sustainable Development

(iv) Resource Utilization as per Carrying Capacity:
If the carrying capacity of a system is crossed (say, by over exploitation of a resource), environmental degradation starts and continues till it reaches a point of no return.

Carrying capacity has two basic components:
i. Supporting capacity i.e. the capacity to regenerate
ii. Assimilative capacity i.e. the capacity to tolerate different stresses.

(v) Improving Quality of Life Including Social, Cultural and Economic Dimensions:
Ensuring the distribution of fare share of benefits

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Sustainability and Sustainable Development

Environmental Stewardship

Environmental Stewardship

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Environmental Stewardship

Introducing Environmental Stewardship

Stewardship of the environment refers to **protecting the environment** through **recycling, conservation, regeneration, and restoration**.

It means **taking responsibility for our choices**.
The responsibility for environmental quality should be shared by all those whose actions affect the environment.

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Environmental Stewardship



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Environmental Stewardship

Defining Environmental Stewardship

Individual or Organizations practice:
✓ Use resources efficiently
✓ Reduce waste and minimize on the negative impacts on the environment
✓ Translate reduced consumption and waste into financial savings
✓ Provide a healthy, more efficient and effective working environment

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Environmental Stewardship

Defining Environmental Stewardship

Embodies the concept of a Green Office, a Green School or Green Home which is conceptualized as smarter and better as it is:
❑ Ecological - using non-toxic, recycled, environmentally friendly products and supplies
❑ Efficient – using as little energy and other resources as possible and putting out the smallest amount of waste as possible
❑ Healthy – will generate as little visual, noise and physical pollution as possible

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Environmental Justice and Equity

What is the Problem?

- The impact of environmental policies and decisions have differentiated impacts across population groups
- Most countries have utterly no idea what environmental justice is nor have any mechanism for addressing EJ issues

Examples

- Haiti and much of the developed Western Hemisphere
- Environmental neglect in New Orleans
- The former East German Republic and its neighbors
- Philippines Islands and Japan
- Sub-Saharan Africa and Europe

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Environmental Justice and Equity

National Policy Alternatives

- Improving information exchange? Knowledge is power and leverage
- National or state environmental justice legislation or regulation
- Non-governmental action groups or agenda setting

Outcomes and Futures

- EJ resolution cannot solve all the world's social/economic justice problems
- International EJ issues may be the next great world policy issue

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Climate and Climate Change



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Climate and Climate Change

Climate – is the statistics of the environment over time. It can include seasonal cycle, extreme events, year to year variations

Overall weather pattern of precipitation and air temperature in an specific area over time.

Weather – what is happening right now in the atmosphere. It's a combination of Temperature, Humidity, Precipitation, Cloudiness, Visibility, Wind and Atmospheric pressure.

Or

Weather – is the condition of the outside air or atmosphere at a particular time and place

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Climate and Climate Change

Climate change refers to any **significant change** in measures of climate (such as temperature, precipitation, or wind) lasting for an extended period (decades or longer).

- Climate change may result from:
  - natural factors, such as changes in the sun's intensity or slow changes in the Earth's orbit around the sun;
  - natural processes within the climate system (e.g. changes in ocean circulation);
  - human activities that change the atmosphere's composition (e.g. through burning fossil fuels) and the land surface (e.g. deforestation, reforestation, urbanization, desertification, etc.)

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